

EFFECT OF LONG TERM EXPOSURE TO HIGH FREQUENCY RADIO WAVES FROM MOBILE DEVICES ON HUMAN FERTILITY

STATEMENT OF THE PROBLEM

Mobile communication is undoubtedly a great advancement in the world of telecommunication from the early 1980s. Unlike the land line, this form of communication doesn't require physical connection between sender and receiver and allows them to change physical location during communication, which makes it more versatile. It uses radio waves to carry signal from a user to a base station rather than traditional cables. Over the past decades, mobile communication has kept on evolving from the early voice-based analog technology 1G, we transitioned to second generation and third generation 3G of voice, SMS text and data service with further increased speed in the 4G generation. The recent 5G uses 5G new radio technology that forms the backbone of numerous applications like VR/AR, IOT, and cloud-based services. This evolution is accompanied with increased use of mobile devices that emit high frequency radio waves, currently estimated at 18 billion subscribers worldwide, which has raised concern over possible health effects on the human body as we are constantly exposed to these high frequency radio waves from mobile phones and cell towers scattered across geographical locations.

Studies from most of the previous research have pointed out that there is no adverse effect on health from radio frequency emitted from mobile devices, but also they pointed out that there is not enough information to conclude that they pose no risk (FDA - 1999). Some studies, however, have suggested the existence of biological effects like changing in brain activity, reaction time, and sleep pattern, which WHO pointed out as small effects and has no apparent health significance (WHO - 2000). Though it is well known that high power radio frequency can burn tissue, the effect on low powered mobile devices is not much conclusive and still ongoing. However, most of these studies are based on short term exposure of the entire body and not long term exposure of specific body parts. This study therefore will specifically focus on the effect of long term exposure to radio waves from mobile devices on human fertility through epidemiological studies.

REFERENCES

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